

Research Briefing| UK

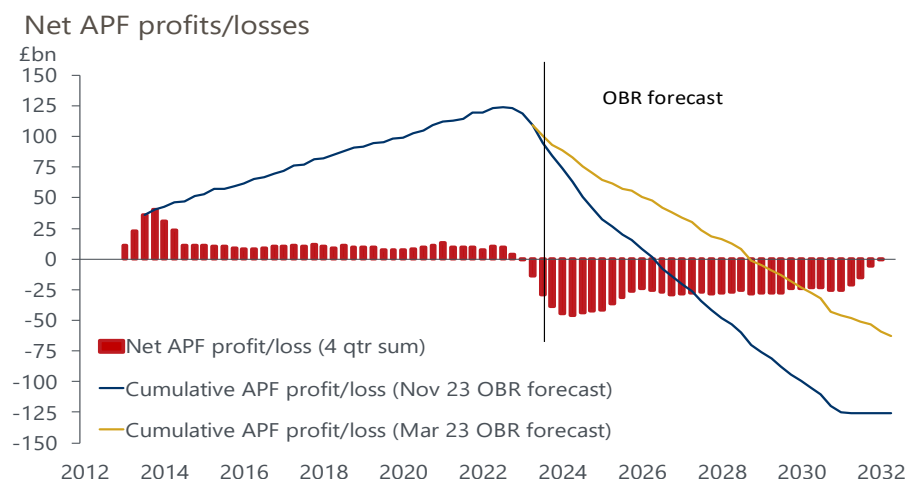
Rethinking QE in light of large asset purchase losses

- Recent official projections suggest that the Bank of England's asset purchase facility – which holds bonds bought under the bank's quantitative easing program – will incur substantial losses due to higher interest rates and bond yields since those assets were purchased.
- The APF's losses are likely to be much bigger than those from other central banks' asset purchase programs, reflecting the large scale and long maturity of the BoE's asset purchases. These losses are also likely to be much larger than expected a few years ago.
- The way these losses interact with the government's fiscal rules will have undesirable side effects. In our view it would be better to exclude the effects of APF transfers from the debt measure used in the fiscal rules. The APF losses will cause public debt to rise. But the appropriate fiscal response is to gradually reduce debt levels outside periods of crisis rather than (as might be required under the current fiscal rules) to immediately and fully offset APF losses with more fiscal consolidation.
- The evidence that asset purchase programs can have a large fiscal cost calls for a rethink of policy options if interest rates again fall to the effective lower bound. It would probably be more cost-effective to restrict future asset purchase programs to short and medium-dated bonds, rather than also include long-dated bonds which are generating a lot of the APF losses.
- We also think the fiscal rules should be reformed to allow more scope for public investment to support activity if interest rates again are at the ELB, rather than rely solely on QE.

The [Office for Budget Responsibility recently revised up its projection](#) for transfers from the government to the Bank of England's asset purchase facility to cover losses during the period of APF rundown (from late-2022 to mid-2031). It now estimates transfers of about £250bn (9.2% of this year's GDP), up from £187bn in March this year. Allowing for net transfers from the APF to the government of £124bn up to mid-2022, the OBR estimates imply the lifetime cost of quantitative easing will be about £126bn – twice the estimate from last March, and equal to 14% of the peak stock of asset purchases (£895bn).

Why are the APF's losses so large and so much bigger than the losses on central bank asset purchases in other countries? Here, we examine the root causes and, in light of evidence that the fiscal consequences of quantitative easing are much more adverse than previously expected, we offer suggestions for rethinking the fiscal rules and policy options when interest rates are at the effective lower bound.

Chart 1: The BoE's QE programs now seem likely to incur a substantial fiscal cost over their lifetime



Sources: Oxford Economics/Haver Analytics

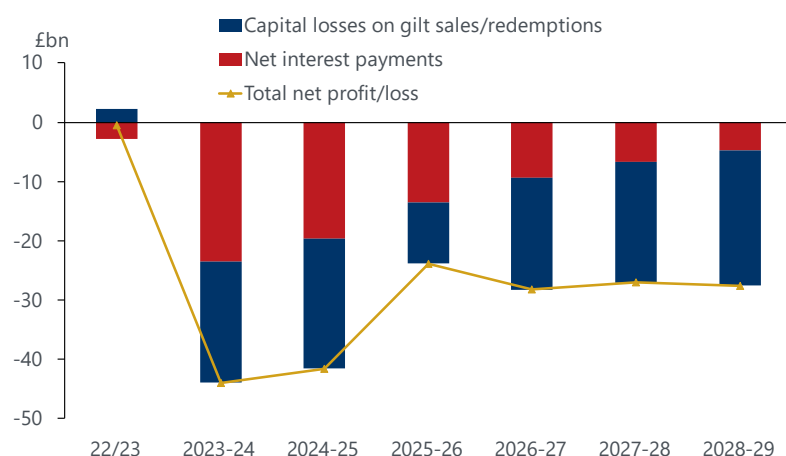
Where do the APF losses come from?

The bonds – mostly gilts – bought through the BoE's asset purchase programs are held in the asset purchase facility. The APF in effect is a large leveraged fund, with assets of £744bn of gilts financed at Bank Rate. Since 2012, any profits or losses have been transferred to or from the Treasury every quarter. Over the period from 2012 to end-2021, the APF made a cumulative profit of £124bn, because the yield on its assets (around 2%) was above Bank Rate. Since then, it has made losses of about £30bn, because of the higher level of Bank Rate and capital losses on bond sales and redemptions.

The OBR projects APF losses will total about £250bn (9.2% of 2023 GDP) from Q4-2022 to mid-2031 when, at the current rate, the APF rundown will be completed. Of this £250bn, roughly one third (£80-90bn) is expected to come from net interest payments, reflecting the fact that Bank Rate and its forward path has risen well above the APF's interest income. The remaining two thirds (£160-170bn) reflects capital losses, as gilts are sold (or redeemed) at below their purchase price. The OBR projects that net interest losses will shrink over time as the stock of gilts falls, whereas capital losses will be fairly steady from year to year and will account for most of the APF's ongoing losses four or five years ahead and after.

Chart 2: Capital losses on gilt sales and redemptions will account for most APF losses in coming years

OBR Projections of annual APF profit/loss



Sources: Oxford Economics / Haver Analytics

Transfers from the government to cover the APF's losses add directly to the public debt measure used for the fiscal rule. On the OBR's forecasts, these losses will add 8ppts-9ppts to the debt ratio between 2022 and 2031. The APF's surplus or shortfall on net interest income also counts towards the fiscal deficit.

The rise in projected APF losses since the March Budget reflects the higher market path for Bank Rate, plus greater capital losses because gilt prices have fallen. The faster pace of APF rundown announced by the Monetary Policy Committee in September has brought forward some losses that would previously have been projected to occur in later years, but it does not alter the net present value of those losses.

The eventual gains and losses from the asset purchase programs may differ from current estimates (in either direction) depending on swings in Bank Rate and gilt prices. The recent drop in market interest rate expectations, if sustained, will reduce losses somewhat. Conversely, [BoE estimates from November](#) using slightly different assumptions for yields and the pace of the APF unwind suggest that APF losses will exceed the OBR's estimate and total about £300bn from 2022 to the early 2030s.

Either way, APF losses currently seem likely to be much higher than expected a few years ago, reflecting the sharp rise in interest rates since then. In mid-2021, the [OBR projected](#) that APF earnings during 2021/2022 to 2031/2032 would range from a £20bn loss to large surpluses under various scenarios. Forecast losses over that period now exceed £200bn.

By any measure, these are large sums. The lifetime cost of QE (£126bn on the OBR's estimate) is similar to the costs of the full HS2 scheme (see [here](#), adjusted to 2023 prices).

The BoE's APF losses are much larger than those on other central banks' asset purchases

Profits from asset purchases in the period before 2022 were probably similar across advanced economies, because policy rates everywhere were very low and below yields on purchased assets. Other central banks have also indicated that they will face losses on asset purchases for similar reasons as the UK. Comparisons are not straightforward because of different accounting policies. But the OBR's forecast of APF losses from 2022 onwards (£250bn, 9% of annual GDP) and the fiscal spillovers appear likely to be much larger than in other countries.

Like the BoE, eurozone central banks and the US Fed include realised capital gains and losses in their profit figures when bonds are sold or redeemed, and do not include unrealised gains and losses. Unlike the UK, any profits/losses on their asset holdings are partly offset by other profits/losses made by the central bank (or accumulated profits), and so net transfers to/from the governments do not equal net profits/losses on asset purchases.

[IMF estimates published in July](#) suggest that losses on asset purchases will cause the eurozone central banks to record losses of about €55bn during 2023 and 2024 combined (about 0.5% of annual GDP), but that these losses will be fully recouped through a recovery in profits by 2027. Within that, the IMF projects peak cumulative losses of almost 1.2% of GDP for the Bundesbank, 0.7% of GDP for the Banque de France, 0.3% of GDP for De Nederlandsche Bank, 0.2% for Banco de España, and no losses for Banca d'Italia. The [DNB itself](#) has reported that it expects to make cumulative losses of €10bn-11bn over 2022-2028, equal to just over 1% of annual GDP. [Separate estimates](#) suggest the eurozone central banks will incur total losses of €640bn (4.7% of annual GDP) on their asset purchase programs over the period 2023-2034 (the comparable figures for the UK are £250bn, 9.2% of annual GDP, and the UK had a smaller stock of asset purchases).

The [US Fed projected early this year](#) that losses on its SOMA account (which includes asset purchases) will total \$115bn (0.4% of annual GDP) in 2023 and 2024 combined, and that aggregate SOMA profits over 2022-2025 (the period in which the projection assumed a shrinking balance sheet) will be about \$330bn (1.2% of annual GDP) below a normal level. Subsequent research by the [St Louis Fed](#) suggests that aggregate losses may be a little higher. But, even so, they are likely to be well below the APF's losses. The Fed has suspended its regular remittances to the government (typically [0.3%-0.6% of annual GDP](#)), but unlike the UK it does not transfer losses to the government. The studies project the Fed will earn its way back to profitability and resume these remittances in a few years.

The RBA and Riksbank report profits and losses including unrealised losses and gains on existing holdings (unlike the BoE, Fed, and eurozone central banks). The [Riksbank](#) reported losses of SEK85bn over 2020-2022 combined (1.4% of annual GDP) and expects it will [need about SEK80bn](#) (1.3% of annual GDP) in capital from the government. The [RBA](#) reported losses of A\$47bn over the last three years, equal to 1.8% of annual GDP, but projects it will earn its way back to a positive equity position over the next 10 years or so without needing a transfer from the Government.

The only central bank to record (or project) losses on a similar scale to the BoE APF is the SNB, which has very large FX assets (equal to roughly 100% of annual GDP) and (unlike the BoE) includes unrealised gains and losses. The [SNB has reported](#) a cumulative loss of CHF84bn (equal to 10.4% of annual GDP) since the start of 2020, mostly reflecting large unrealised losses on its FX assets. However, the fiscal effects of the SNB's losses are small compared to the BoE APF. The SNB has suspended its usual CHF6bn (0.7% of annual GDP) annual payment to the cantons, but otherwise has absorbed these losses out of existing equity and reserves. Moreover, the SNB's foreign assets recorded substantial gains in prior years, and the cumulative return over the last ten years is, unlike the APF, slightly positive.

Note that these figures do not represent a fully comprehensive measure of the fiscal effects of asset purchases. Such an assessment would also need to include their indirect effects in terms of lower bond yields (cutting government debt service costs) and higher economic growth (cutting benefit outlays and lifting tax revenues). These offsetting positives would probably be similar in other countries and hence would not alter the point that the APF's losses are likely to be relatively large.

Why are the BoE's APF losses so large compared to other countries?

The relatively large size of the UK's APF losses reflects the interplay of several factors.

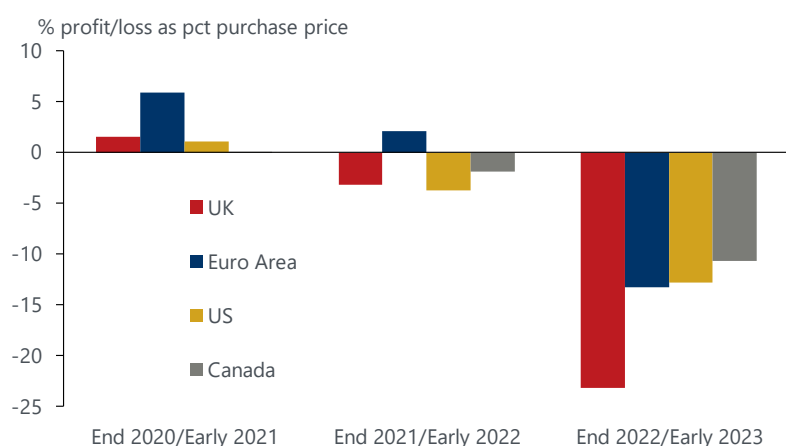
First, the BoE's asset purchase programs were relatively big. The stock of asset purchases peaked at 34% of GDP in Q3-2022, similar to the US (32%) and below the eurozone (52%), but well above equivalent figures for Australia (22%), Canada (14%), and Sweden (14%).

Second, the average maturity (and hence duration) of the APF's bond holdings has been far higher than other AE central banks. Hence, a given rise in interest rates and bond yields would cause bigger losses for the APF on bond sales (or a longer period of negative net interest payments) than for other central banks. [Comparable BIS figures](#) show the average maturity of bond purchases during the pandemic were 13 years for the APF, seven-eight years for the Fed and ECB, around six years for the RBA and BoC, and around four years for the Riksbank. The longer maturity of APF holdings reflects the relatively long average maturity of outstanding gilts, and the BoE's decision to spread gilt purchase across the yield curve.

As a result, in February this year, the APF had an [unrealised mark-to-market loss](#) of 23% (£193bn) on its £830bn bond holdings. This compares to equivalent figures for unrealised mark-to-market losses of 13% for the Fed's domestic bond holdings in June; 13% for the ECB and the six largest eurozone central banks combined at end-2022; and 11% for the Bank of Canada (also end-2022). Aggregate losses on domestic securities (relative to their prior value) over the last three years have been about 11% for the RBA and 7% for the Riksbank.

Chart 3: The BoE APF has reported relatively large mark-to-market losses on its bond portfolio

Mark-to-market profit/loss rate on central bank bond holdings



Sources: Oxford Economics / Haver Analytics

Based on current market prices, we estimate that the average mark-to-market loss for the BoE APF is now still about 23% (£174bn loss on £744bn current holdings). Government bond prices for the other countries have moved similarly, and hence their mark-to-market losses are probably also little changed.

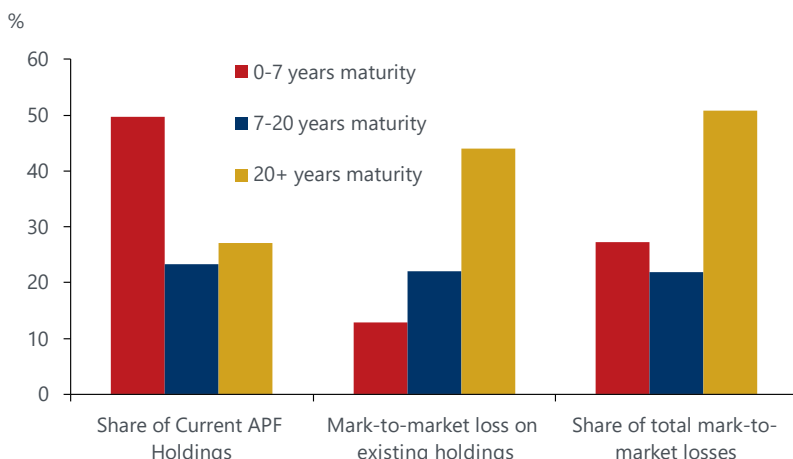
The split of the APF's losses highlights the role of long-dated gilts. So far, the APF has made gilt sales of £59bn (valued at purchase price) and recorded £19.2bn of capital losses – an average loss rate of 33%. Within this, the average loss rate on gilts of maturity 20+ years has averaged 53% (so £23.7bn of long gilts sold has crystallised a £12.5bn loss); they have accounted for 40% of APF gilt sales but 65% of APF capital losses on gilt sales. By contrast, the average loss rate on gilt sales with maturity below seven years has been 16%, and that for 7-20 year maturity gilts has been 24%.

It's likely to be a similar story going forward. Of the APF's remaining gilt holdings, 27% have maturity of 20+ years but at current asset prices, these have an average loss rate of 44% (versus 13% for short-dated gilts and 22% for mediums), and account for just over 50% (about £90bn) of the APF's £174bn mark-to-market losses.

If anything, these figures understate slightly the role of long-dated gilts in generating APF losses. This is because, with the passage of time, some gilts purchased with a maturity over 20 years have migrated to shorter maturity categories. Hence, losses on some gilts purchased a maturity of 20+ years now count as losses on short or medium-dated gilts.

Chart 4: Long-dated gilts account for half the APF's mark-to-market losses on gilt holdings

Split of APF holdings and mark-to-market losses by maturity



Source: Oxford Economics / Haver Analytics

Third, since the BoE APF has an indemnity, any gains/losses on it flow immediately and fully through to the government's finances. This is not the case for most other central banks. Many can offset losses on asset purchases against accumulated reserves. Or, if they have a negative net equity position, they can use other sources of income (including seigniorage from supplying bank notes), or suspend net transfers to the government, to regain a positive net equity position over several years.

Fourth, the BoE has not taken any offsetting action to lift its income. By contrast, the [ECB has cut to zero](#) the interest rate on banks' required reserves. The [SNB made a similar change](#) and also has introduced a lower interest rate on excess reserves above a specified threshold. These measures, even if partly driven by other factors, will have the effect of improving the profitability of those central banks and offsetting losses from asset holdings.

Is it possible to reduce these losses?

The short answer is probably no.

Any attempt to limit APF losses by keeping Bank Rates lower than needed to achieve the inflation target would have highly adverse side effects and destroy the UK's anti-inflation credibility.

Since so much of the APF's losses reflect capital losses on gilt sales, it might be tempting to think these losses could definitely be reduced by holding the gilts until redemption (i.e., passive rundown of the APF) rather than active gilt sales. However, this view is misplaced.

If the BoE was not selling its gilt holdings, the result would (with the current market rate path) simply be a very long period of net interest losses from the APF over coming decades until the last gilt held by the APF is redeemed in 2071. The decision to sell gilts rather than allow passive rundown through redemptions brings those losses forward to the next few years, but does not alter their net present value. To be sure, the Fed is shrinking its balance sheet by passive rundown. But, given the shorter duration of its purchases, its losses would be smaller whether the bonds are sold or held to maturity.

And for the UK, a hold-to-maturity strategy for the APF over several decades would have disadvantages. QE has effectively shortened the maturity of public sector debt. This is an acceptable temporary side-effect of QE's role as a means of stimulus when inflation is below target. But it is undesirable for the long term because it would raise the public finances' sensitivity to changes in short-term interest rates. The UK Debt Management Office and Treasury, which are responsible for optimising the structure of public debt, have for good reason chosen to have long maturity debt to reduce such rollover risk. So, maintaining a large stock of purchased assets indefinitely might also limit the scope for future QE, because further QE would exacerbate this issue. This is less of an issue elsewhere because, with the shorter duration of asset purchases, the stock of asset purchases unwinds relatively quickly once reinvestment of existing holdings ends.

Of course, a hold-to-maturity strategy might turn out to be profitable for the APF if yields happened to fall. But it might just as equally make things worse if yields rise in coming decades. It's not really feasible to expect the APF to successfully adopt a market-timing approach, seeking to sell more gilts when yields are temporarily lower and fewer when yields are higher. The patchy record of active fund managers suggests that such market timing is unlikely to persistently succeed. And, by raising uncertainty over the pace of gilt sales, it may cause an additional risk premium on yields. The DMO has a long-held practice of making gilt sales predictable for these reasons. Likewise, given that APF rundown will (at the current rate) take ten years or so, APF losses would not be significantly different if rundown began a quarter or two earlier or later.

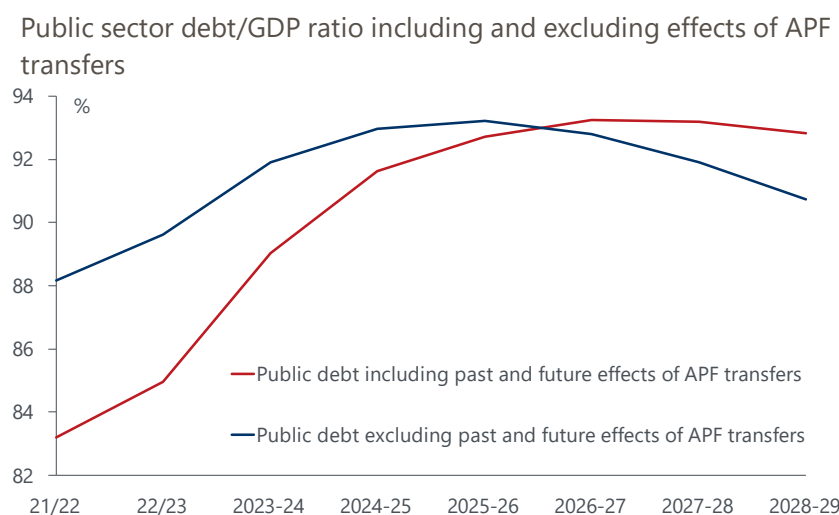
It is of course possible for the BoE, like the ECB and SNB, to implement other changes (e.g., like the SNB and ECB, [paying no interest on part of banks' reserves](#), which is akin to a tax on bank assets) that raise its underlying profits. This would not reduce losses from asset purchases but would offset their fiscal effects. However, since the APF's losses are indemnified by the government, unlike the SNB and ECB, the BoE itself does not have a strong incentive to make such a change.

How should the APF losses interact with the fiscal rules?

At present, the pace of APF rundown has a direct and substantial impact on the path of the public debt ratio from year to year, and hence on the government's ability to meet its key fiscal rule of a falling debt ratio five years ahead.

The OBR projections suggest that transfers to cover APF losses will add an average of 1.1ppts per year to the public debt ratio during 2023/2024 to 2028/2029, a cumulative effect of 6.8ppts, with a further rise of 1ppt-2ppts of GDP after that. But if the MPC were to accelerate the pace of APF rundown so that it all occurs in the next five years, the public debt ratio would rise faster in that period (with no APF effect subsequently), making it much harder for the government to meet its debt target. Conversely, slower APF rundown would make it much easier to achieve the fiscal rules by pushing more APF losses beyond the five-year horizon for the debt rule.

Chart 5: APF losses are likely to add substantially to public debt in the next few years



Sources: Oxford Economics/Haver Analytics

In our view, this strong feedback from MPC actions to the government's ability to hit its fiscal rules is undesirable. In the near term, it creates powerful incentives for the Treasury to influence MPC decisions on the pace of APF rundown in order to meet its fiscal rules, thereby potentially eroding the MPC's independence. The BoE should be able to unwind the stock of asset purchases at whatever rate it regards as optimal to achieve its remit, without having to worry about how this will affect the government's ability to meet its fiscal rules.

Further ahead, the Treasury might be reluctant to sanction future asset purchases, even if these are justified under the MPC's remit, because of the risk that the unwind of those asset purchases might affect the government's ability to meet its fiscal rules in the future.

Moreover, fiscal policy should be guided more by medium-term fiscal sustainability trends rather than such one-off effects.

In our view, the solution is to [exclude the effects of transfers to and from the APF](#) (or to just exclude the effects of capital losses on gilt sales and redemptions) from the public debt measure used for the fiscal rules. This would be consistent with the approach during the pandemic and global financial crisis when the fiscal rules were suspended to allow the necessary support for the economy in such extraordinary circumstances. The APF losses largely reflect asset purchases made during those crises and represent a delayed cost of those crises. They do cause a rise in public debt. But the appropriate fiscal response is to steadily reduce debt levels outside such crises – i.e., fix the roof when the sun is shining – rather than as might be required under the current fiscal rules immediately and fully offset APF losses with additional fiscal consolidation.

This issue does not really arise in other countries because central bank losses will be smaller, do not immediately affect the government's finances, or because governments, such as the US and Canada, do not have fiscal rules like those in the UK and EU countries.

Is QE the most cost-effective form of stimulus at the ELB?

If Bank Rate is at the effective lower bound and the economy needs more stimulus to prevent persistent below-target inflation, then in theory either fiscal policy or QE could be used. An increase in public investment lifts demand directly, but QE lifts demand indirectly by [lowering longer-term yields](#), for example through portfolio balance effects, signalling that policy rates are likely to remain low, and reducing liquidity premia on eligible assets. All this is likely to boost growth by reducing borrowing costs for households and businesses, depreciating the exchange rate, and lifting asset prices.

However, comparing the costs and effects of these options is not straightforward because QE's effects on the economy and its potential costs are both uncertain. Our assessment below sketches out an approach.

A [broad analysis](#) suggests that for the UK, a QE program of 1% of GDP (currently about £27bn) typically has a peak effect on GDP of 0.12%, and a long run effect typically after five or seven years of only 0.05% – virtually nothing. BoE studies on average have suggested a similar peak effect but a long run effect that is even closer to zero. Using the expected cost of the existing QE programs as a guide, a QE program of 10% of GDP would have a lifetime cost of about 1.4% of GDP. Of course, this is not guaranteed.

The [OBR's fiscal multipliers](#) suggest that an increase of public investment of 1.4% of GDP would have a peak effect on GDP of about 1.4% (i.e. an initial multiplier of about 1), but would also fade over time. This is broadly similar to the possible costs and effects of a QE program of 10% of GDP.

The effects on activity of fiscal policy and asset purchases depend partly on conditions at the time. The OBR's fiscal multipliers reflect the typical impact over periods when stronger growth would prompt higher interest rates. However, fiscal multipliers are probably much larger when interest rates are at the ELB and inflation is low, because offsetting monetary tightening would be less likely (see [here](#) and [here](#)). This is the relevant comparison because QE would only be used under those conditions. At such times, therefore, fiscal policy – or fiscal policy combined with QE – might be a more cost-effective tool for stimulus than QE alone. QE's effect on yields and hence probably on growth also varies: it is typically [bigger in times of financial stress](#) and smaller at other times.

We have also run simulations on our in-house model. In one, we assume a 50bp drop in bond yields, which, based on these [BoE estimates](#), is roughly what might be expected from a QE program of 10% of GDP at times when financial markets are not stressed. In the other, public investment rises by 1.4% of GDP. Both options might be expected to have a similar cost, based on the existing QE programs. We assume Bank Rate does not respond. Under these assumptions, public investment again appears to be a more cost-effective form of stimulus than QE.

Certainly, a full assessment would have to allow for the greater uncertainties around the costs and effects of asset purchases. But it's unlikely that relying solely on QE will always be the optimal response when monetary policy is at the ELB and the economy needs more stimulus.

At present, the UK policy framework is not well designed to consider whether asset purchases are likely to be the most cost-effective option available. There is an "underlap", with neither the BoE or Treasury having clear responsibility for this judgement.

The UK's fiscal rules, which require the government to aim to reduce the debt and deficit over time, prevent fiscal policy from playing a role in stabilising the economy other than when the rules are suspended, such as during the GFC and pandemic. Of course, any decision to expand QE needs Treasury approval because it carries an indemnity. But it's unclear to what extent the Treasury assesses the possible fiscal impact of the MPC's QE proposals during this approval process and compare it against alternatives. The deadline for approval is typically a few days at most, so is too short to allow much analysis. So far, no request to expand QE has been rejected. And if the Treasury were to reject QE, it would leave the MPC in a very awkward position.

At the same time, in judging whether to use asset purchases to support growth, the MPC does not view consideration of the fiscal effects of those asset purchases as its responsibility. This was recently made clear by [BoE Deputy Governor Dave Ramsden](#) "But importantly, the MPC does not take into account financial risk or profit when taking monetary policy decisions, including about the APF gilt portfolio." If Bank Rate is at the ELB, and the MPC judges that more stimulus is needed to achieve its remit of returning inflation to target, then the MPC's only option is to expand QE.

It appears that the Treasury and BoE have shared an implicit assumption that QE's fiscal effects would be negligible, or sufficiently small that they do not warrant much attention, and hence QE can be treated purely as a monetary policy tool. That may have seemed a reasonable assumption when QE began but in light of experience seems more questionable now.

A better approach would be first, to acknowledge that the long run fiscal effects of asset purchases need to be considered in advance and, second, to [reform the fiscal rules](#) by introducing an escape clause that allows more fiscal flexibility when monetary policy is constrained by the ELB. A possible model for such circumstances would be to put less emphasis on reducing public debt and more on lifting public sector net worth aims, while allowing a multi-year return to balance on the current budget.

These changes would allow the government, perhaps in conjunction with the BoE, to consider the potential costs and benefits between asset purchases, fiscal policy, and other policy tools as options for stimulus at the ELB. Since gilt yields would typically be very low when the policy rate is at the ELB, these changes would probably allow a significant rise in public investment, which could help lift the economy out of the ELB trap while also yielding supply-side gains. These changes would allow consideration of other quasi-fiscal monetary policy options for stimulus, for example a [dual interest rate scheme](#) whereby the central bank provides funding to banks at below Bank Rate to reduce bank lending rates. If, under those conditions, the government chose not to use fiscal policy or other measures, then by default it would make QE more likely.

Is it possible to re-design QE to be more cost effective?

Considerations of cost-effectiveness should prompt a rethink in the design of any future QE programs.

So far, the UK's asset purchase programs have been spread across the whole yield curve, roughly reflecting the distribution of government debt and taking some account of factors such as liquidity. Underlying this has been an implicit assumption that what matters is the total scale of asset purchases and their effect on the overall yield curve, and that any cost considerations are not relevant.

However, the potential cost/benefit effects from QE are probably not even across the yield curve.

The potential costs of QE are (as seen) much higher for purchases of long-dated gilts. If held to maturity with an adverse running yield, long gilts create a net interest cost extending decades ahead. If sold, the scope for capital losses is far higher than for short-dated gilts, because the price of long gilts is typically more sensitive to yield changes. As noted, so far, the average loss rate on sales of long gilts (20+ years) has been slightly more than 2x that on mediums and more than 3x that on shorts (53%, 24%, and 16% respectively). Of course, losses on long gilts may not be so high in any future asset purchase programs. But they will always be more uncertain, and potentially much larger, than those for short and medium-dated gilts.

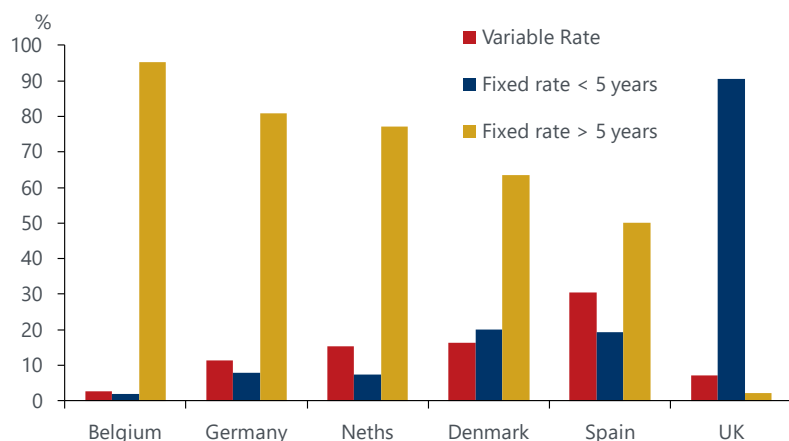
Conversely, the likely effects on the economy probably are no smaller (and may be bigger) if asset purchases are concentrated on bonds with maturities up to 15-20 years rather than long-dated gilts.

The effects of QE through asset prices and the exchange rate are driven by the overall level of yields along the curve, including long yields. However, QE's effect on borrowing costs for households and businesses through portfolio rebalancing would probably be greater at short and medium maturities. Most corporate bank debt is floating rate or short maturity fixed rates (below five years), although some companies have longer term bond debt. Most mortgages have fixed rates with maturity of five years or less, and, unlike the US and many eurozone countries, few have fixed rates over 10 years. It is conceivable that changes in 30–40-year yields might affect demand by changing expectations of households and firms for borrowing costs in future decades. But this is probably less than the economic impact from changing the borrowing costs that households and firms actually face at present, which are driven mainly by shorter-term rates.

Likewise, QE's signalling channel on the likely path of interest rates in the next few years is probably more powerful if asset purchases are concentrated at maturities below 10 years, because that is the horizon for the relevant policy signal. It's unlikely that asset purchases of 40-year bonds convey a credible signal for policy rates, either in the next few years or over the next 40 years.

Chart 6: Long-term fixed-rate mortgages account for a small share of UK mortgage debt

Split of new mortgages over last five years



Sources: Oxford Economics / Haver Analytics

Given this, it's unlikely that purchases of long gilts, which currently have an average loss rate of around 50%, provide a cost-effective form of stimulus. It would make sense to concentrate any future asset purchase programs on the shorter part of the yield curve, perhaps out to 15-20 years or so. This would allow asset purchases to have the same traction as now on yields in this part of the curve, which covers the most relevant maturities for private sector borrowing costs, while limiting potential fiscal costs. If, for example, the APF currently held only gilts with maturity of 15 years or less (i.e., purchases had excluded longs) then at current yields, QE's lifetime costs would be small – a point also made [here](#).

One way to achieve this without compromising the MPC's independence in setting monetary policy would be for the Chancellor to include a requirement in the MPC's remit for any future QE programs to be designed to be as cost-effective as possible. Or the Treasury could simply insist, as part of any future indemnity for asset purchases, that QE concentrates on short and medium maturities.

Rethinking QE can expand the range of policy options and make QE more cost-effective

The likelihood that the BoE asset purchase programs will have a large overall cost does not necessarily mean that the decisions to launch those asset purchase programs were wrong. At those times, the MPC had no other options to stimulate the economy, it was unclear whether fiscal policy would be sufficiently counter-cyclical, and the possible adverse fiscal effects of asset purchases were not widely appreciated.

The possibility of renewed asset purchases may seem distant right now, but in our view they shouldn't be dismissed. The [neutral level of interest rates](#) is probably still relatively low, and the lurching crises of the last 20 years suggest that we may face major irregular shocks that bring the ELB back into play.

As a result, even with the knowledge that QE can have large fiscal effects, there may still be a role for asset purchases in the future. Like other monetary policy options, QE has the key advantage that it can be deployed very quickly and at large scale if needed, whereas fiscal policy typically is more slow-moving. QE may still be the most appropriate tool in circumstances, like those in early 2020, when a sharp rise in the demand for money causes gilt yields to rise rapidly, hence producing a sharp and undesirable tightening in financial conditions. At other times, even a low pace of asset purchases may have a useful role as a signal that interest rates will stay low for a while.

However, it's important to learn broader lessons to ensure any future QE programs are as cost-effective as possible. If asset purchases are urgently needed during a crisis, those purchases could be rapidly scaled back – and unwound – as the crisis eases. And greater consideration could be given to other forms of stimulus if Bank Rate is at the ELB, such as fiscal policy or quasi-fiscal tools.